SAMPLE

History of Astronomy: Egyptian Astronomy

Stars of the Nile

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Introduction

This volume belongs to the History of Astronomy shelf, the room of the library that follows the longest human conversation with the night. The subject here being the astronomy of ancient Egypt, the five-thousand-year tradition that ran from the cattle-keepers' stones of the desert to the medieval observatories of Cairo, and the book's business being to tell that tradition whole: the calendars and the star clocks, the alignments and the coffin ceilings, the river that kept the time and the sky that kept the record.

The room's arrangement being familiar to readers of the companion volumes: one continuous telling, no apparatus of citations, no imported encyclopedia text, the sums left visible and the claims kept inside what the monuments and the papyri can carry. The spine of dates being the standard one: the Nabta Playa stones of the fifth millennium before the common era, the pyramids of the Old Kingdom, the star tables of the Middle Kingdom coffins, the tomb ceilings of the New Kingdom, the Greek merge at Alexandria, the Roman eclipse, and the Cairo of Ibn Yunus a thousand years later. Any reader wanting to check the load-bearing facts against scholarship will find the scholarship named at the back of the volume.

The tradition the dates describe being both stranger and more familiar than the modern reader expects: stranger, because the Egyptian sky was a goddess whose body was the night and because the dead were timed by clocks painted inside their coffins; familiar, because the same tradition produced the three-hundred-sixty-five-day year the whole planet still uses, the twenty-four hours that divide every modern day, and the habit of checking the calendar against the sky that no civilization has since abandoned.

The chapters taking the room's usual route: discovery first, then understanding, characteristics, science, observation, features, context, the modern verdict, research, discoveries, challenges, the future, trivia, and finally significance. One thread carrying the whole telling: the river having set the problem, the sky having supplied the answer, and the Egyptians having built between the two the longest-running astronomical institution the species has ever maintained.

The book's two debts being declared at the door: the first to the Egyptologists who spent the last two centuries turning the inscriptions back into astronomy, the second to the modern surveyors whose theodolites confirmed what the temples always claimed about their own axes. The volume having drawn on both trades freely and admitting the drawing, the admission being the shelf's own discipline, and the reader being the beneficiary of a tradition that happens to be unusually well preserved, the preservation being the mortuary habit's own gift to the future.

One preliminary clarification saving the reader a common confusion: the Egyptian astronomy of this volume being the native tradition of the Nile, the Greek astronomy of Alexandria belonging to a later merged chapter of the same long story, and the medieval Cairo astronomy belonging to the tradition's Islamic-era afterlife, all three being told here because the country hosted all three and the shelf refusing to cut the story at the dynastic border. The volume's own emphasis being the native tradition: the calendars, the tables, the alignments, and the ceilings that the pharaonic state produced, the production being the main event and the later merges being the epilogue the history requires.

The reader being offered the volume's own small legend before the tour begins: the Egyptian word for hour having been a star's job title, the hour's name having been the name of the duty rather than the clock, and the legend being the whole volume in one etymology, the etymology being the reader's own key to every chapter that follows. The book's sixteen sections being sixteen hours in the ancient reckoning, the reckoning being the subject's own invention, and the invention being the reason the modern reader's day being divided into pieces this volume can occupy one at a time, the occupying being the reader's own choice and the choice being the last thing the door says before the tour starts.

The introduction owing the reader one more honesty: the volume being a synthesis rather than an excavation, the reader being walked through what two centuries of scholarship have established rather than through the raw texts, and the synthesis having been checked against the standard editions at every load-bearing point. The reader wanting the raw material having been given the pointers, the pointers being the shelf's own courtesy, and the courtesy being the volume's last promise before the walking starts.

Chapter 1: The Discovery of Egyptian Astronomy

The discovery beginning in the desert, millennia before the first pyramid. The stone circles of Nabta Playa having been raised in the fifth millennium before the common era by cattle-herding people in what is now the southern Egyptian desert, the largest circle having carried a line of sight to the point where the midsummer sun rises, and the site having included cattle burials beneath stones that dwarf the graves. The place having been a ceremonial center standing a thousand years before the better-known megaliths of the far north, and the calendar-keepers of the Sahara having watched the same June sun their remote descendants would greet at Karnak.

The calendrical discovery being the tradition's foundation: the three-hundred-sixty-five-day civil year having been in use at the very beginning of recorded Egyptian history, the year having been built from three seasons of four thirty-day months plus five festival days kept outside the months, and the seasons having been the river's own: the inundation, the growing, and the harvest. The year being the calendar's whole reason: the Nile flood arriving on a schedule the state had to know in advance, and the sky having been the state's almanac long before it was anyone's science.

The pyramid discovery belonging to the Old Kingdom: the great tombs of Giza having been set to the cardinal directions with an accuracy the modern surveyor can measure and admire, the north having been found in an era when the pole star was not the modern one, the faint star Thuban of the Dragon having stood near the celestial pole in the pyramid age, and the builders having aimed their corridors and their kingdom at that slow pivot. The descending and ascending passages of the Great Pyramid having been aimed with equal care, the southern shafts of the king's chamber pointing to the region of Sirius and the belt of Orion, the north shafts to the sky's turning point, and the alignments being the earliest large-scale astronomical engineering that survives anywhere.

The temple discovery completing the set: the great temple of Amun at Karnak having been laid out on the midwinter sunrise, the axis having been built so the year's lowest sun rose in line with the sanctuary's door, and the pattern having been repeated at other temples of the Nile. The midsummer sunrise having been honored at Nabta Playa five millennia ago and the midwinter sunrise at Karnak two millennia later, and the two alignments bracketing the tradition's whole instinct: the sun's turning points having been the land's fixed addresses from the first.

The textual discovery arriving with the coffins: the diagonal star tables, painted on the inside surfaces of wooden coffin lids from about the ninth and tenth dynasties onward, the tables being grids of thirty-six star names crossed with the twelve divisions of the night, and the grids having told the dead, hour by hour, which stars would serve as the night's clock. The tables being the oldest surviving astronomical documents of the Nile, and the dead having been the tradition's best customers, the afterlife having been imagined as a night to be navigated, and the star tables having been the traveler's timetable.

The discovery's full arc closing with the two merges: the Greek settlement of Alexandria having made the country the ancient world's scientific capital, the native tradition having flowed into the Greek and Babylonian streams there, and the medieval astronomers of Cairo having kept the observation of the sky alive into the era of the astrolabe. The discovery of the tradition by the moderns being the chapter's last movement: the hieroglyphs having yielded their astronomy in the nineteenth century of the common era, and the modern field of archaeoastronomy owing to the Nile one of its richest archives.

The discovery's textual wing deserving its own paragraph: the star clocks having been joined by the ceiling texts of the New Kingdom tombs, the royal burial chambers having carried the sky goddess painted across the stone vault with the sun's evening and morning journeys inscribed beside her, and the ceilings having been the tradition's picture books. The best-preserved ceilings having been counted, catalogued, and in the modern era photographed entire, the photographs being the discovery's own modern sequel, and the sequel having been running since the hieroglyphs first yielded their vowels to the philologists.

The late discovery of the Cairo era completing the arc: the astronomers of the medieval city having kept the Nile's observing tradition alive into the era of precision instruments, the tenth-century master of the age having watched the sun and the eclipses with a great astrolabe for decades, and his eclipse records having been consulted by the astronomers of later centuries, the consultation being the tradition's own long echo. The arc from the desert stones to the medieval astrolabes being the discovery chapter's whole span, the span being five millennia and the chapter having walked it in a page.

The discovery's modern movement deserving the fuller telling: the hieroglyphic script having kept the astronomical texts sealed for fifteen centuries, the decipherment of the nineteenth century having opened the archive, and the twentieth century having produced the great editions that turned the ceilings and the tables into a working corpus.

The reader being reminded that the tradition was discovered twice, once by the Egyptians who built it and once by the moderns who read it, the two discoveries being separated by the long silence of the lost script, and the silence having been broken within the last two hundred years, the years being recent and the archive being ancient, the contrast being the discovery chapter's own closing thought.

The pyramid discovery's astronomical interior deserving the fuller paragraph: the Great Pyramid's internal shafts having been aimed at the sky's significant addresses, the king's chamber having received two shafts toward the north and two toward the south, the northern pair having pointed to the region of the pole and the southern pair to the region of the hunter's belt and the fleet star, and the aims being deliberate in the modern analysis, the analysis having been conducted with the computed sky of the pyramid age. The precision being the paragraph's exhibit: the builders having cut shafts through a hundred feet of stone to catch the light of particular stars for one dead king, the undertaking being the largest astronomical instrument the ancient world ever aimed at a single person, the person having been the state, and the state having been aimed at the sky.

The discovery closing with the name of the modern field: the archaeoastronomy of the Nile being a named discipline with its own conferences, its survey instruments, and its debates, the discipline having been assembled in living memory, and the assembly being the discovery chapter's own coda: the tradition having been discovered, lost, rediscovered, and institutionalized, the four stages being the full biography of every ancient science and the Egyptian case being the biography's best-told volume.